

Project Management Plan

PICKLINK - Report 2

Project Name	PickLink-Pickleball Community Connection Platform
Vision & Scope Ref	Report 1
RTW Ref	[ProjectName]_RTW.xlsx (Requirements Traceability Workbook)
Plan Version	v1.0.0
Date Created	21/05/2026
Last Updated	26/05/2026
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Status	Draft

1. Overview

1.1 Cost/Time Estimation

- Total hours = 2,520
- Total days = 105 days = 15 weeks = ~3.5 months
- Efforts include analyzing, designing, coding, and testing.
- We do at least 5 days/week and maybe all week:
Maximum man-days capacity of team = 7 (days) × 15 (weeks) × 5 (people) = 525
- Estimated man-days capacity of team = ~370

#	Work Package	Est. Effort (pds)	Deadline
Stage 1: Project Initiating (Week 1)			
1.1	Report 1 (Project Introduction)		Week 1
Stage 2: Project Planning & Initial Requirements (Weeks 2–3)			
2.1	Report 2: Project Management Plan v1.0		Week 2
2.2	Identify the business process((BPMN model)		Week 2
2.3	Report 3: SRS v0.9 (Overall Requirement)		Week 2
2.	Vertical Prototypes (Code Demo)		Week 3
Stage 3: Software Design (Weeks 4–5)			
3.1	Vertical Prototypes (POC — Proof of Concepts)		Week 4
3.2	Report 4: SDS v1.0 (High-Level Design Specifications)		Week 4
3.3	Report 5: Test Planning		Week 5
3.4	Report 3: SRS v1.0 (Detailed Requirements for Iteration 1)		Week 5
Stage 4: Implementation (Weeks 6–11)			
4.1 Implementation Iteration 1 (Weeks 6–7)			
4.1.1	Report 4: SDS v1.1 (Detailed Design Specifications)		Week 6
4.1.2	Code Package & Unit Testing Report		Week 6
4.1.3	Report 5: Integration Test Cases		Week 6
4.1.4	Report 5: Integration Test Report		Week 7
4.1.5	Report 3: SRS v1.1 (Detailed Requirements for Iteration 2)		Week 7
4.1.6	Software Package Version 1		Week 7
4.2 Implementation Iteration 2 (Weeks 8–9)			
4.2.1	Report 4: SDS v1.2 (Detailed Design Specifications)		Week 8
4.2.2	Code Package & Unit Testing Report		Week 8
4.2.3	Report 5: Integration Test Cases		Week 8
4.2.4	Report 5: Integration Test Report		Week 9
4.2.5	Report 3: SRS v1.2 (Detailed Requirements for Iteration 3)		Week 9
4.2.6	Software Package Version 2		Week 9
4.3 Implementation Iteration 3 (Weeks 10–11)			

#	Work Package	Est. Effort (pds)	Deadline
4.3.1	Report 4: SDS v1.3 (Detailed Design Specifications)		Week 10
4.3.2	Code Package & Unit Testing Report		Week 10
4.3.3	Report 5: Integration Test Cases		Week 10
4.3.4	Report 5: Integration Test Report		Week 11
4.3.5	Report 5: System Test Cases		Week 11
4.3.6	Software Package Version 3		Week 11
Stage 5: Verification & Validation (Weeks 12–13)			
5.1	Report 5: System Test Report		Week 12
5.2	Report 6: User Guides (Installation Guide, User Manual)		Week 12
5.3	Acceptance Test and Support		Week 13
5.4	Final Software Package (Documents, Source Codes, etc.)		Week 13
Stage 6: Closing (Week 14)			
6.1	Report 7: Final Project Report		Week 14
6.2	Presentation File		Week 14

1.2 Project Objectives

State committed objectives with planned vs. actual metrics. Update actuals at project close.

#	Metric	Unit	Planned	Actual	Notes / References
1	Effort Usage	Person-day	370		Total allocated effort for the project.
2	Number of Defects	No. of defects	175		Total expected defects across all testing stages (Reviewing: 20, Unit Test: 60, Integration Test: 40, System Test: 40, Acceptance Test: 15).
3	Timeliness	%	100%		Commitment to achieve 100% Milestone Timeliness.
4	Requirement Completeness	%	100%		Ensure all team members fully complete assigned tasks and system requirements.
5	Test Coverage	%	100%		Committed test coverage across all test stages (Reviewing, Unit Test, Integration, System, Acceptance).
6	Learning & Technology Adaptation	-	100%		Qualitative objective: All team members learn new knowledge and adopt new technologies during execution.

1.3 Project Risks

List the top risks (up to 5).

Provide both mitigation (prevent) and contingency (react) plans; Priority = Impact × Possibility (H/M/L).

#	Risk Description	Impact	Possibility	Priority	Response Plans (Mitigation / Contingency)
1	Requirement change: Scope is subject to change; Some requirements may be changed, added or removed from time to time.	High	High	High	All members discuss carefully at the start of each sprint to define the scope and requirements. Make sure that resource allocations are changed correctly to accommodate the new requirements.
2	Conflict among team members: Team members disagree with others and refuse to work or work below their capacity.	High	High	High	Set clear and consistent tasks before starting the task. All members discuss conflict resolution. If members cannot verbally reconcile, the team will proceed to vote.
3	Sickness or absence of team members, in particular. The number of members according to personal schedules/working time in the company, decreased and not enough for work.	High	Low	Medium	Team members should keep the team informed of their absences, individual schedules, and follow-up plans for work. Ensure that the absence of one member will not affect the others and always have a plan to deal with this. The project Manager reminds team members about task deadlines.
4	Team member distraction: Team members are not engaged enough in work time.	Medium	Medium	Medium	Choose members carefully. The PM reminded the team members about the schedule.
5	Lack of skill & technology skill	High	Medium	High	Members must acquire new skills or technologies to develop the system. Members can chat to decide which features should be out of scope.

2. Management Approach

2.1 Project Processes

Development Methodology: The project will follow an **Agile/Scrum** methodology

Scope Management: All features must be documented as User Stories in the Product Backlog.

Change Management: New feature requests or scope changes identified mid-sprint are added to the Product Backlog for future prioritization.

Time Management: Time is managed through daily 15-minute Stand-up meetings to track progress and identify bottlenecks.

Issue Management: Bugs and technical blockers are logged as "Issues" in the project management tool

2.2 Quality Management

Defect Prevention: Emphasis on clear Acceptance Criteria for every User Story and utilizing Clean Architecture principles to decouple business logic from framework dependencies.

Peer Reviews: Mandatory Pull Request (PR) reviews. Code cannot be merged into the **main** or **develop** branch without approval from at least one other team member.

Unit Testing: Focus on writing unit tests for core business logic, specifically the ELO calculation algorithm, matchmaking logic, and financial/cost-splitting calculations.

Code Standards: The team will adhere to standard linting rules (e.g., standard Dart guidelines for mobile, Microsoft C# conventions for backend).

2.3 Project Training Plan

Plan training for any skill gaps identified in the team.

Training Area	Participants	When / Duration
ASP.NET Core & Clean Architecture	Backend Team	2 days (week 3)
Git & GitHub	Entire Team	1 days (week 2)
Agile/Scrum Workflow	Entire Team	1 days (week 2)
React Fundamental	Frontend Team	2 days (week 3)

3. Responsibility Assignments

Describe the project organization and assign RACI roles for each work package listed in Section 1.1.

Work Package	ThinhTHHE 163873	DatNVHE 180346	AnhDNQHE 180752	NhatNQHE 181012	NamNTPHE 181708
<i>R = Responsible A = Accountable C = Consulted I = Informed</i>					
Project Planning & Tracking	R	R	R/A	R	R
Prepare Project Introduction Document	R	R	A	R	R
Prepare SRS Document	R	R	A	R	R
Prepare SDD Document	R	R	A	R	R
Code backend	C	R	C	R	R
Code frontend	C	R	R	C	C
Prepare Test Document and Report	R	C	R	R	R
Prepare User Guides	R	R	R	R	R
Prepare Final Report (all document and presentation slide)	R	R	R	R	R

4. Project Communications

Define recurring communication events, who owns them, and how they are conducted.

Communication Item	Who / Target	Owner	Purpose	When / Frequency	Type / Tool
Daily Meeting	All team members	Quang Anh	Check the progress of assigned tasks. Assign new tasks to team members.	10 PM every day	Online meetings
Weekly Meeting	All team members	Quang Anh	Provide	Usually on weekends	Online meetings
Meeting with Supervisor	All team members, Supervisor	Quang Anh	Check the progress of the project. Discuss existing problems and new requirements	Once a week in Slot 4 on Wednesdays at school.	Offline meetings
Emergency Meeting	All team members	Quang Anh	Discuss and solve critical problems	When critical problems occur.	Online meetings

5. Configuration Management

5.1 Document Management

All project documents are centrally stored and managed to ensure consistency and easy access:

Storage: Google Drive (Shared Workspace). A root folder named "ĐỒ ÁN SU26" is created and shared among all team members.

Access Control: Team members are granted "Editor" access for collaborative drafting. The Supervisor/Reviewer will be granted "Viewer" or "Commenter" access to monitor progress and provide feedback without altering the original content.

Naming Convention: Document files must follow the syntax:

[ProjectCode]_[DocumentName]_v[Version]_[DateUpdated].docx

(e.g., *PickLink_Report_2_ProjectPlan_v1.0_21052026.docx*).

Version Management: The built-in "Version History" feature of Google Docs is utilized to track individual contributions and changes. Upon reaching a major milestone, a PDF version will be downloaded (to lock the content) and uploaded to the official repository folder.

5.2 Source Code Management

The source code for the PickLink system will be managed using a distributed version control system (Git).

Repository: GitHub

Branching Strategy: A standard **Git Flow** model is applied to prevent code conflicts:

- **main:** The most stable branch, containing only tested and defense-ready code.
- **develop:** The primary development branch where code from feature branches is integrated.

- **feature/[feature-name]**: Branches created from **develop** for individuals to work on specific features independently (e.g., **feature/matchmaking**, **feature/booking**).
- **bugfix/[bug-name]**: Branches used for fixing bugs identified during testing.

Commit Convention: Commit messages must clearly follow the Conventional Commits standard: **[Type]: [Short description in English]**. (e.g., *feat: Add ELO calculation logic* or *fix: Resolve database connection timeout*).

Code Review Process: Developers are strictly prohibited from pushing code directly to the **main** or **develop** branches. Instead, a **Pull Request (PR)** must be created. At least one other team member must review the code and approve the PR before it can be merged into the target branch.

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Technology	ReactJS (Front-End), ASP.NET Core Web API (.NET 8) (Back-End), Entity Framework Core 8 (ORM)
Database	SQL Server & SQL Server Management Studio (SSMS)
IDEs / Editors	Visual Studio 2022 Community
Diagramming	Draw.io, Visual Paradigm Community Edition, Canva
Documentation	MS Office version 365, Google Docs / Sheets / Slides
Version Control	GitHub (Source Codes), Google Drive (Documents)
Deployment Server	Azure App Service, SmarterASP.NET
Project Management	GitHub Projects, ProjectLibre (scheduling), GitLab (tasks, time logs, defects, issues)
API Testing & Docs	Postman, Swagger